

Flipped Classroom Exercise Set
NEUROSCI 366, Brain Function Through the Lens of Computation
Discussed: Friday, September 19, 2025

1 Operations with vectors and matrices

Computational neuroscience requires familiarity with vectors and matrices. These mathematical objects are typically introduced in high school algebra and studied deeply in a college linear algebra course. This problem serves to remind students of these fundamental concepts and demonstrate the corresponding MATLAB commands. NOTE: Wikipedia articles and MATLAB help pages are great resources for mathematics and programming.

Consider the following scalars, vectors, and matrices,

$$s = 2, u = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, v = \begin{pmatrix} 2 \\ 5 \end{pmatrix}, w = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 5 \end{pmatrix}, A = \begin{pmatrix} 0 & 0.5 \\ 10 & 1 \\ 2 & 9 \\ 2 & 7 \end{pmatrix}, B = \begin{pmatrix} 4 & 4 \\ 8 & 9 \end{pmatrix}, C = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}. \quad (1)$$

- What is the general rule from mathematics that determines when scalars, vectors, and/or matrices can be summed with each other? Which pairs of mathematical objects in Eq. 1 can be summed? Evaluate these sums with MATLAB. Warning - MATLAB will not complain if you try to sum s with any of the other variables. How does MATLAB interpret $s + A$, and does this operation correspond to the abstract sum of a scalar and a matrix? What does it correspond to?

NOTE: In this and subsequent bullets, please interpret “pairs” to mean two different mathematical objects. For instance, you should consider pairing u with A but not with itself.

- Which pairs of vectors in Eq. 1 can be combined through an inner product? Evaluate the result by hand and in MATLAB.
- What is geometrical meaning of a vector’s norm? How does the norm of a vector relate to the inner product? Evaluate the norm of w , denoted $\|w\|$, with MATLAB.
- Which pairs of vectors in Eq. 1 can be combined through an outer product? How many distinct mathematical objects can be produced by taking outer products between these pairs of vectors?
- What is the general rule from mathematics that determines when a matrix can multiply a vector? What is the dimensionality of the resulting vector? Which pairs of matrices and vectors in Eq. 1 can be combined through a matrix product? Choose one such pair and evaluate the matrix product by hand and in MATLAB.
- What is the general rule from mathematics that determines when two matrices can be multiplied together? What is the size of the resulting matrix? Which pairs of matrices in Eq. 1 can be combined through a matrix product? Choose one such pair and evaluate the matrix product by hand and in MATLAB.

- Explain the concept of an identity matrix. Write down an example of one, and generate it in MATLAB. Note that an identity matrix is usually denoted as I .
- What is the definition of the matrix transpose? Evaluate the transpose of A , denoted A^T , in MATLAB. Which vectors in Eq. 1 can A^T multiply?
- By interpreting an N -dimensional vector as an $N \times 1$ matrix, the transpose of a vector can be interpreted as a $1 \times N$ matrix, termed a row vector. Write the norm of w in terms of w and w^T .
- What is the definition of the matrix inverse? Explain why A is not invertible. Evaluate the inverse of B , denoted B^{-1} , in MATLAB. Note that C is not invertible because the second row is a scalar multiple of the first, and a matrix is only invertible when its rows and columns are linearly independent. What happens if you try to evaluate its inverse in MATLAB?
- For which mathematical objects in Eq. 1 is the determinant mathematically defined? Evaluate these determinants by hand and in MATLAB. What condition on the determinant determines whether a matrix is invertible?
- The Hadamard product is a mathematical operation that combines same-size vectors or matrices by component-wise multiplication. It has little significance to pure mathematics, but it is extremely useful in programming. Which pairs of mathematical objects in Eq. 1 can be combined with a Hadamard product? Evaluate these Hadamard products with MATLAB.

2 Random processes and probability

2.1 Discrete variables - Bernoulli distribution

The Bernoulli distribution describes a simple binary random process that takes the value 1 with probability p and the value 0 with probability $q = 1 - p$. It is called a discrete probability distribution because it can only take values from a discrete set of outcomes (i.e., $\{0, 1\}$). For instance, you can think of this random process as a biased coin flip where “heads” corresponds to the value 1 and “tails” corresponds to the value 0. What is the probability mass function and cumulative density function of the Bernoulli distribution? Use the probability mass function to derive the mean and variance of the Bernoulli distribution as a function of p . What is the standard deviation of the Bernoulli distribution?

2.2 Continuous variables. Exponential distribution

Some probabilistic processes take continuous values. For example, the exponential distribution assigns non-zero probability to any non-negative value according to the probability density function

$$P(x) = \lambda e^{-\lambda x} \quad (2)$$

for $x \geq 0$, where λ is a positive parameter called the rate. Evaluate the cumulative density function. What is the probability that x takes a value in the interval (a, b) for $0 \leq a \leq b$? Calculate the mean, median, mode, and standard deviation of the exponential distribution.

3 Multivariate probability distributions

Multivariate probability distributions allow one to consider the joint outcome of multiple noisy measurements. For instance, a neuroscientist might measure the number of action potentials emitted by many different neurons on the same trial.

3.1 Bivariate normal distribution

The multivariate normal distribution is a very important example that will allow us to illustrate many fundamental concepts. Specializing to the case of two continuous variables, x_1 and x_2 , the probability density function of the bivariate normal distribution is

$$P(x_1, x_2) = P(x) = \frac{1}{\sqrt{(2\pi)^2 \det C}} e^{-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)}, \quad (3)$$

where we've combined x_1 and x_2 into a 2-dimensional vector $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$ is a 2-dimensional vector of means, C is a 2×2 covariance matrix, $C = \begin{pmatrix} \sigma_1^2 & c\sigma_1\sigma_2 \\ c\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$, and $\det C$ is the determinant of C . The bivariate normal distribution has five parameters, μ_1 , μ_2 , σ_1 , σ_2 , and c ? Explain the meaning of each of these parameters.

3.2 Marginal distributions

Without doing a new calculation, what do you think will be the probability density function for x_1 alone if x_1 and x_2 are bivariate normal? In mathematical terms, this is called the marginal distribution of x_1 . It can generally be computed from the joint distribution via $P(x_1) = \int_{-\infty}^{\infty} dx_2 P(x_1, x_2)$.

3.3 Statistical independence

We say that x_1 and x_2 are statistically independent if $P(x_1, x_2) = P(x_1)P(x_2)$. Assuming that x_1 and x_2 are bivariate normal, when are they statistically independent?

3.4 Conditional distributions

It is often useful to compute the probability distribution of one variable under the condition that the value of another has some known value. This is called a conditional probability distribution, and it can be computed as $P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)}$. In general, what is the conditional probability distribution of x_1 on x_2 if x_1 and x_2 are statistically independent? Consider a bivariate normal distribution with $\mu_1 = 0$, $\mu_2 = 1$, $\sigma_1 = 0.5$, $\sigma_2 = 1$, and $c = 0.7$. Is the mean of $P(x_1|x_2 = 0.75)$ larger or smaller than the mean of $P(x_1)$. Why?

3.5 Bayes rule

Use the definition of the conditional probability distribution to write $P(x_2|x_1)$ in terms of $P(x_1)$, $P(x_2)$, and $P(x_1|x_2)$. This simple equation is known as Bayes rule or Bayes theorem.